

Implicit behavioural assumptions in budgeting decisions

Dr. Shashwat Pande
KTP Associate, Data Science
s.m.pande@lboro.ac.uk

Research Context



Research Context

- Innovate UK Knowledge Transfer Partnership (KTP) – Triumph Motorcycles Ltd.
- Largest UK designer and manufacturer of motorcycles (very cool!).
- *'Data Science: Operations and Intelligence'*
 - a) Development of cases for commercial applications of predictive analytics.
 - b) Managerial responses to the adoption of predictive analytics tools, models and frameworks.

Research Context

- Joint work at the CIM, Loughborough University:



Prof. Peter Kawalek

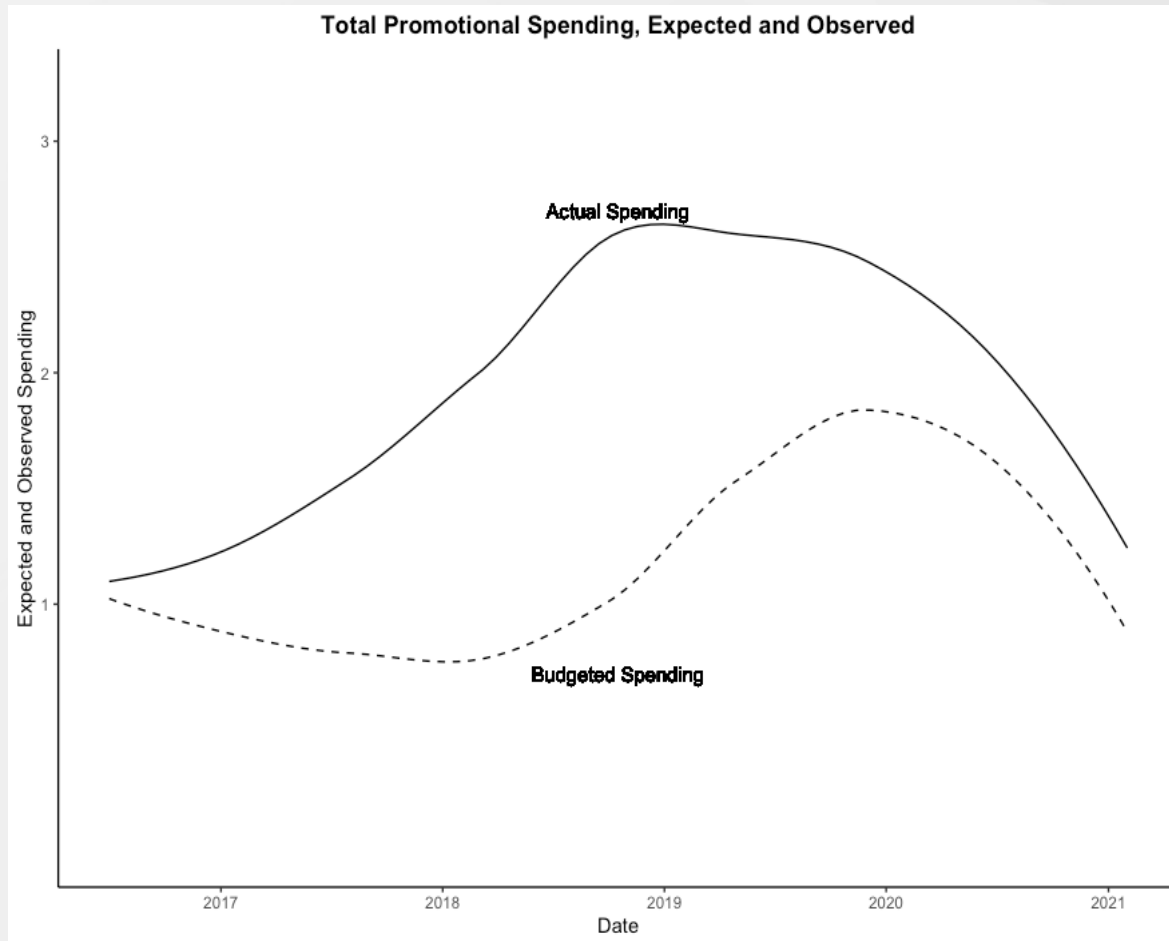


Prof. Christopher Holland

Why is promotional spending underestimated?



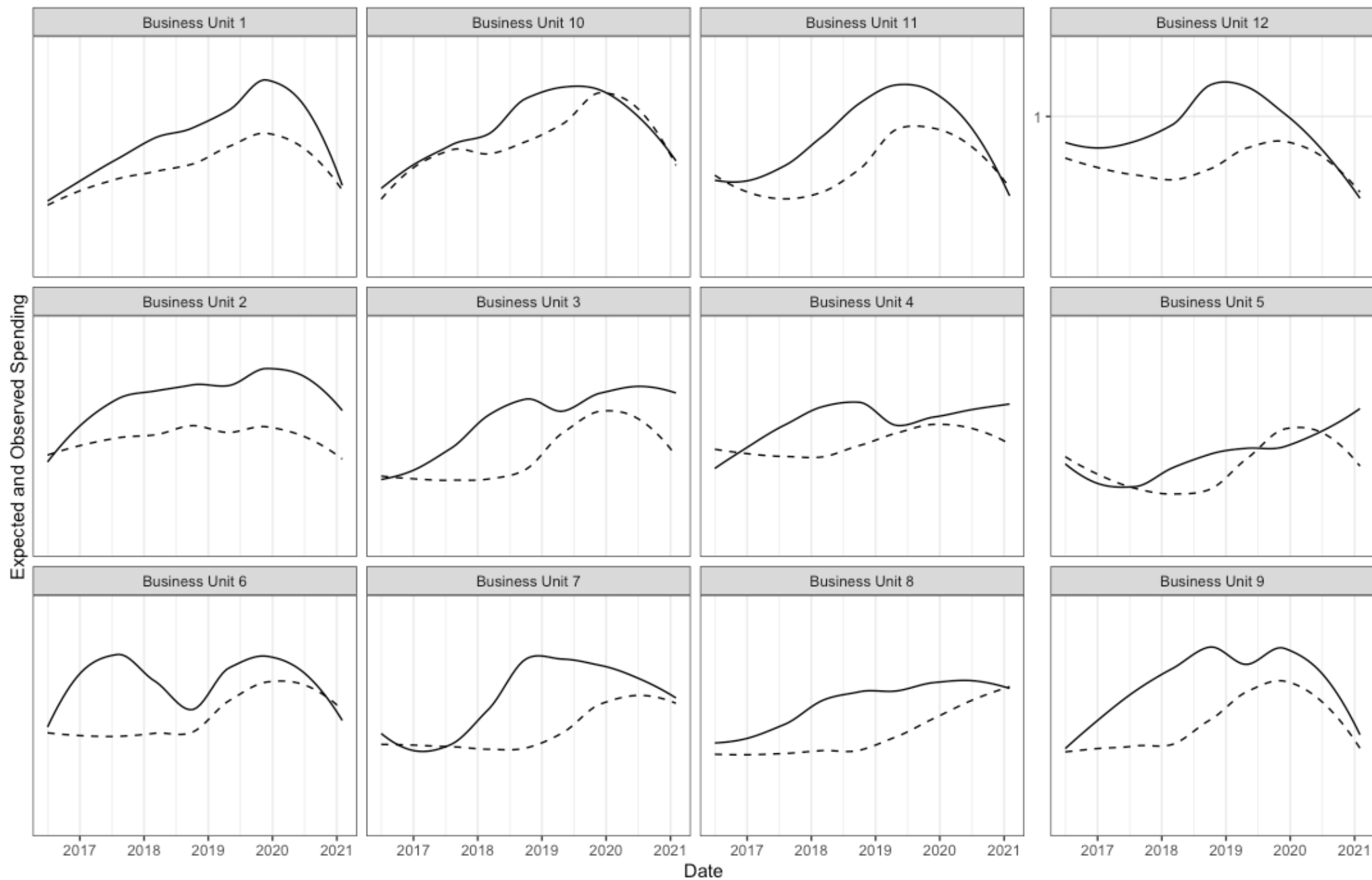
Why is promotional spending underestimated?



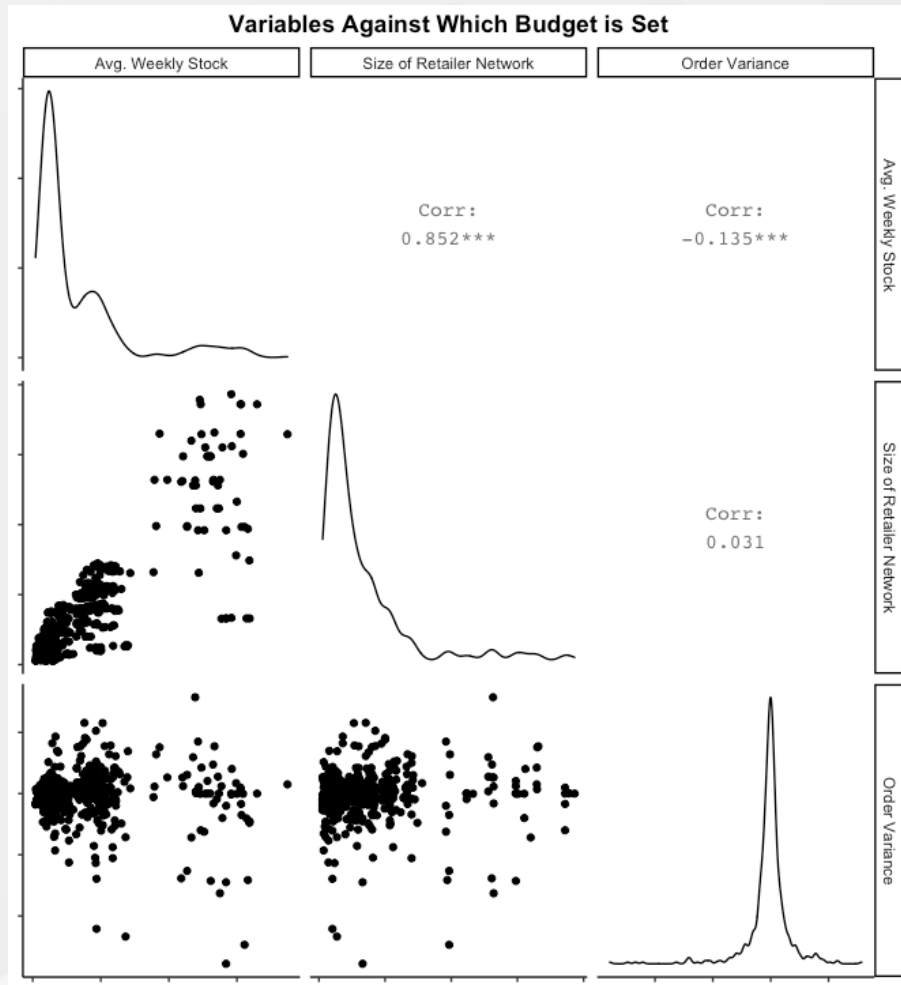
- 5-years of historical data ($n = 672$).
- Across 12 geographically distributed business units ($N = 56$).



Total Promotional Spending, Expected and Observed, by Geographical Business Unit

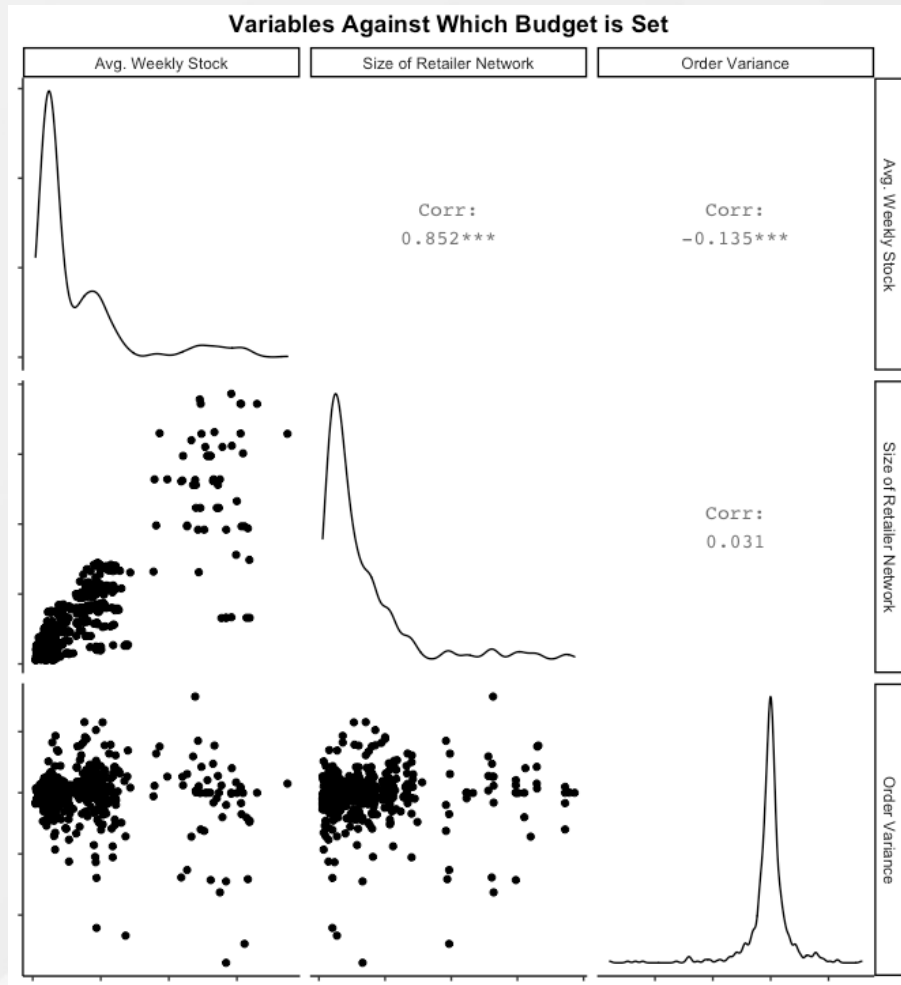


Attributes Considered in Budget Setting



- Informal procedures guide budget setting.
- Based on 3 reports:
 - **Stock:** Avg. weekly units.
 - **Retailer Network:** No. of dealerships.
 - **Order Variance:** [Actual - Planned Orders].

Attributes Considered in Budget Setting



- Intuitive hypotheses:

- + Stock $\Rightarrow$ + Budget

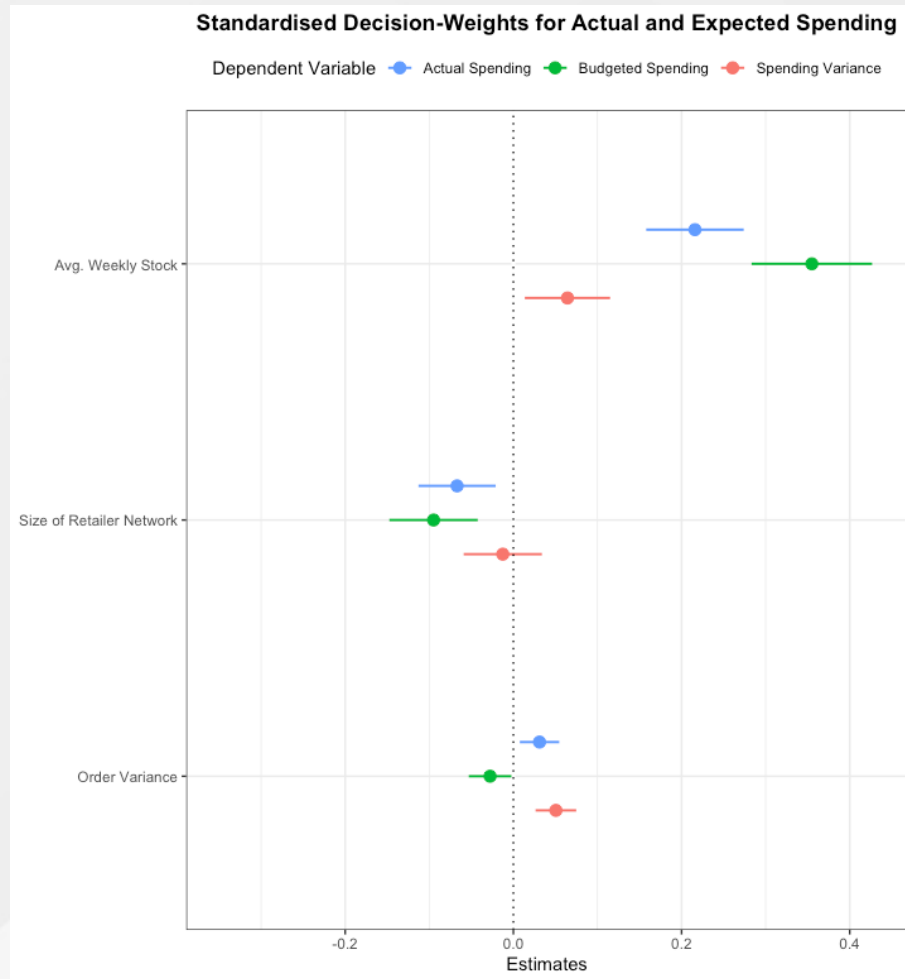
- + Retailers $\Rightarrow$ - Budget

- + Order Variance $\Rightarrow$ - Budget

Estimating Empirical Decision Weights

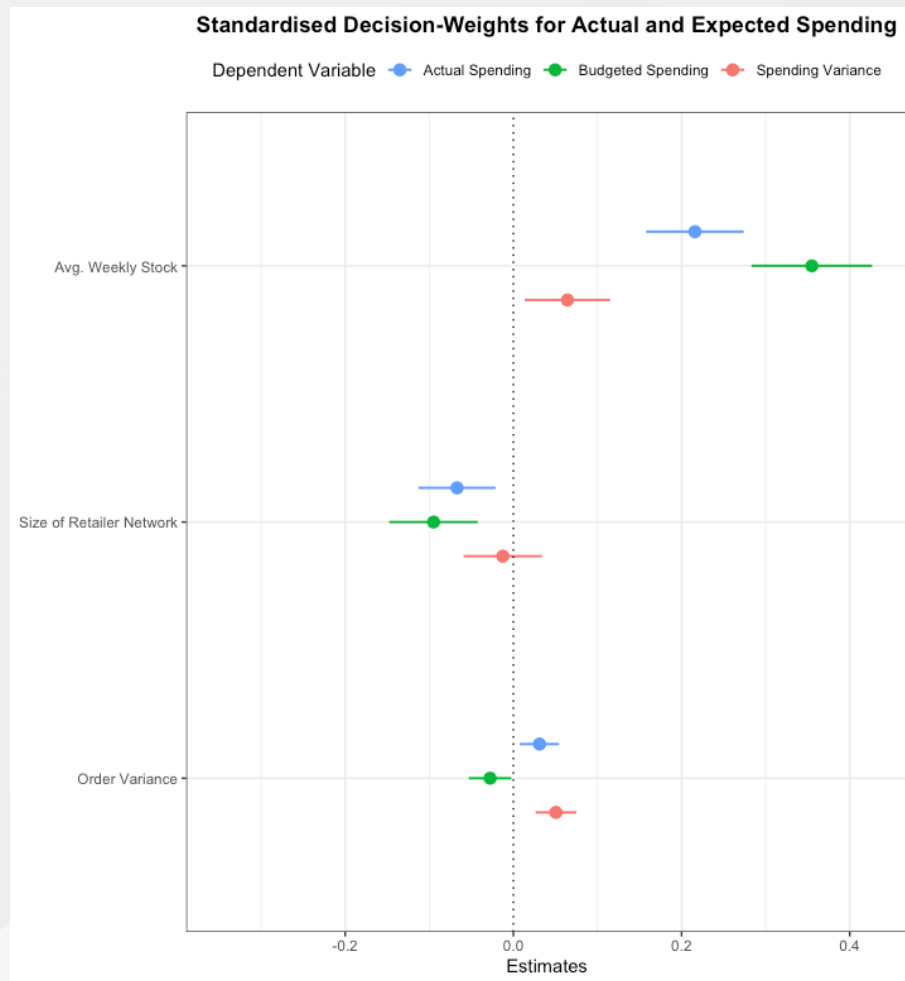
- $Spend_{i,j} = \beta_0 + \beta_1 \cdot Stock_{i,j} + \beta_2 \cdot Retailers_{i,j} + \beta_3 \cdot OrderVariance + \dots + \epsilon$
- Estimation of decision-weights through partial pooling.
- Provides a basis to compare standardised-weights across budgeted and actual spending.
 - Why is promotional spending underestimated?

Preliminary Findings



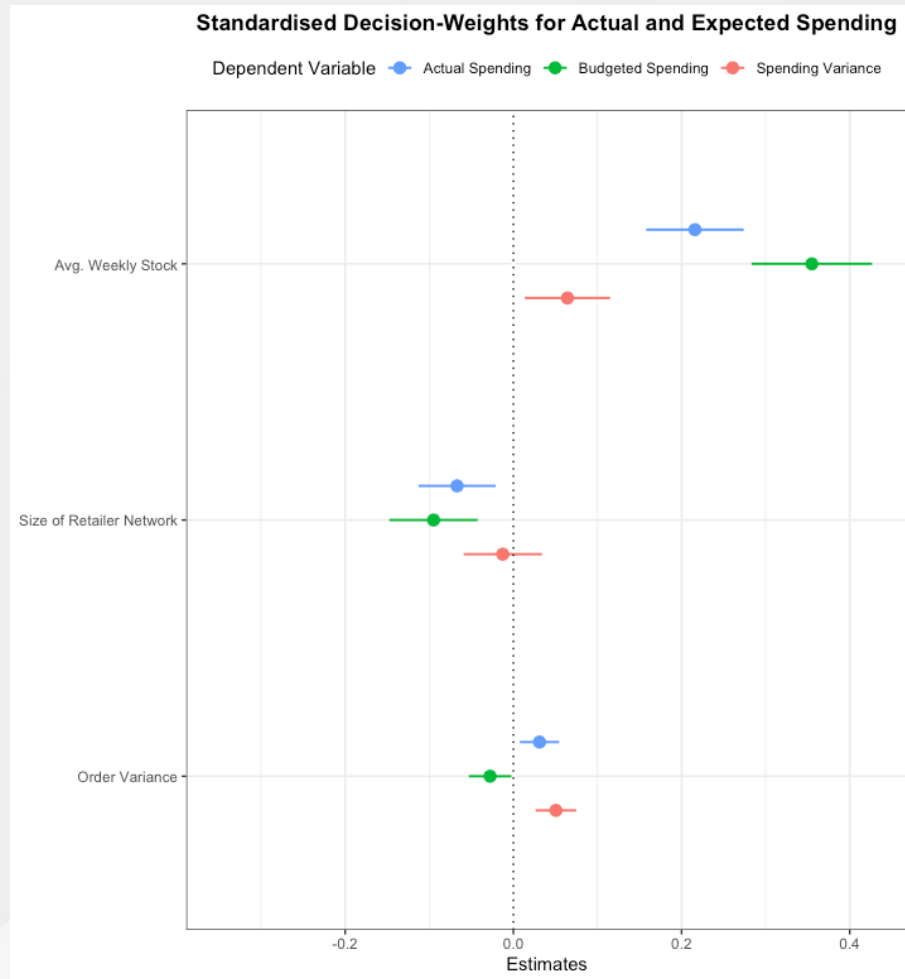
- Overestimation of the importance of available stock.
- Marginal underestimation for the role of the retailer network.

Preliminary Findings



- The role of the 'Order Variance' appears puzzling at first.
- The direction of the effect changes from the budget to actual.

Preliminary Findings



- Despite a relatively small effect-size, its contribution to the variance in spending (Actual – Budgeted) is considerable.
- Why?!

Limitations and Further Research

- A very simple case limited to a single organization.
- **Proposition 1:** While heuristics can be powerful tools, a reliance on informal judgements can lead to assumptions about the behaviour of others which may be problematic in practice.



Limitations and Further Research

- **Proposition 2:** Drawing from the descriptive theory can help contextualise model outputs by assisting in the identification of possible changes to organizational processes through the identification of relevant stakeholder groups.

Thank you for your attention!

Dr. Shashwat Pande
KTP Associate, Data Science
S.M.Pande@lboro.ac.uk



Loughborough
University

#InspiringWinners since 1909